INSTITUTO FEDERAL
Paraíba
Campus Campina Grande

ENGENHARIA DE COMPUTAÇÃO

Circuitos Eletroeletrônicos

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Teoria de controle

Circuito amplificador operacional para resolver equações diferenciais

Simulação no Falstad.com

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<cir f="1" ts="0.000005" ic="382.76258214399064" cb="69" pb="50" vr="5" mts="5e-11">
  <a x="240 208 384 208" f="0" ma="15" mi="-15" ga="100000"/>
  <w x="240 128 240 192" f="0"/>
  <w x="384 208 384 128" f="0"/>
  <r x="240 128 384 128" f="0" r="10000"/>
  <r x="144 128 240 128" f="0" r="10000"/>
  <r x="240 224 240 272" f="0" r="10000"/>
  <g x="240 272 240 320" f="0"/>
  <r x="240 224 112 224" f="0" r="10000"/>
  <a x="560 80 704 80" f="0" ma="15" mi="-15" ga="100000"/>
  <w x="560 32 560 64" f="0"/>
  <w x="704 32 704 80" f="0"/>
  <r x="448 64 560 64" f="0" r="10000"/>
  <g x="560 96 560 112" f="0"/>
  <a x="560 224 704 224" f="0" ma="15" mi="-15" ga="100000"/>
  <w x="560 160 560 208" f="0"/>
  <w x="704 160 704 224" f="0"/>
  <g x="560 240 560 256" f="0"/>
  <c x="448 368 496 368" f="0" c="0.00000999999999999999" iv="0.001" sr="0" vd
    ="-0.018752120317212 16"/>
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  <w x="560 320 560 368" f="0"/>
  <w x="704 320 704 384" f="0"/>
  <g x="560 400 560 416" f="0"/>
  <O x="1056 208 1120 208" f="0" sc="0"/>
  <r x="800 192 864 192" f="0" r="10000"/>
  <r x="864 112 800 112" f="0" r="10000"/>
  <w x="864 192 864 224" f="0"/>
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  <w x="912 192 864 192" f="0"/>
  <g x="912 224 912 240" f="0"/>
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  <w x="1056 208 1056 128" f="0"/>
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  <a x="912 208 1056 208" f="0" ma="15" mi="-15" ga="100000"/>
```

```
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  <w x="864 272 864 224" f="0"/>
  <w x="448 64 448 144" f="0"/>
  <w x="448 368 448 208" f="0"/>
  <w x="384 208 448 208" f="0"/>
  <w x="448 144 448 208" f="0"/>
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  <w x="800 192 704 224" f="0"/>
  <w x="800 272 704 384" f="0"/>
  <w x="1056 208 1056 448" f="0"/>
  <w x="1056 448 144 448" f="0"/>
  <w x="-16 448 -16 128" f="0"/>
  <R x="112 224 64 224" f="16" wf="2" fr="1" maxv="5"/>
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  <r x="640 160 704 160" f="0" r="100"/>
  <r x="496 368 560 368" f="0" r="100"/>
  <w x="144 128 -16 128" f="0"/>
  <w x="144 448 -16 448" f="0"/>
  <r x="448 208 560 208" f="0" r="991"/>
  <r x="560 320 704 320" f="0" r="7723"/>
  <r x="560 32 704 32" f="0" r="10090"/>
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    <p v="0" sc="5"/>
    <p v="3" sc="0.0001"/>
  </o>
  <o en="22" sp="512" f="x2002" p="1">
    <p v="0" sc="5.567959713387315"/>
  </o>
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  <adj e="51" ei="0" en="Resistance (ohms)" mn="100" mx="10000" st="Integral"/>
  <adj e="52" ei="0" en="Resistance (ohms)" mn="100" mx="10000" st="Derivativo"/>
</cir>
```

Listing 1 – Código XML do circuito eletrônico desenvolvido no Falstad

Resultado obtido

